

[G0R72A] Data Visualisation Project Report

“Exploration”

Part 3. Visual design

In this section, you present the visual design phase that you went through to arrive at the designs to be implemented. Do not consider if you will be able to actually implement these designs or not in Part 4: act as if you have a very experienced programmer who would do that for you.

Please provide the following:

1. A description of the process how you created the final designs (which approach, how did you collaborate, ...)
2. Five to 10 sketches (a mix of diverge, emerge and converge sketches) that you think best illustrate the path you took from diverge to emerge and converge. Explain their encoding and why you chose them.
3. Which 2-3 reworked sketches would you want to have implemented if you had an experienced programmer at your disposal? Explain the visual encoding and (if relevant) the interaction. Describe how these designs could help in exploring the general questions mentioned in Part 2, point 3.

You may include a link to an external resource such as Miro, Figma or Excalidraw, but that link is only supplementary: no points will be added/removed if you do or do not add it. Important: paste the full URL in the document instead of a link that automatically opens the web browser (i.e., <http://www.example.com> instead of [example](#)).

Max 3 pages.

Part 4. Implementation

In this section, you describe 3 of the implementations that you created to answer the questions in the brief. Limit yourself to three; if you want you can include additional ones in the screencast. These implementations may be of designs that come directly from part 3 (diverge, emerge or converge), or of new designs. For each visualisation, explain the visual encoding and the interactions: how to interpret the marks and channels in the chart. Include screenshots (with annotations if relevant). You do not have to describe additional plots that you made in e.g. R or Excel to answer the questions; this is about the custom visualisations.

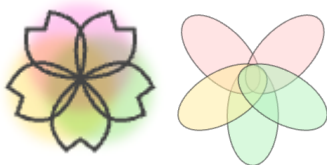
In the description below, we make a distinction between the intended/ideal design, and the actual design. The ideal design is what you want to have in the end (if you had the help from an experienced programmer); the actual design is what you ended up with. See figure below.

Please provide the following **for each custom implemented visual**:

1. The intended design that you wanted to implement: a picture (hand drawn or a mock-up) of the design that you aimed for with the implementation. These could be from the diverge-emerge-converge stage earlier, or new ones. Important: they should be relevant for the questions mentioned in Section 2 of the report.
2. The actual design that you were able to implement: a description of the visual encoding of the implemented visual (what does each part refer to; what are the marks and channels). An annotated image/screenshot may be helpful.
3. A description of the interactions that are possible.
4. If the implementation does not have all features of the drawn sketch (in other words: if the actual implementation is not the same as the ideal one): how does your implementation relate to the target design? What is not included yet, or what needs to be added still if you had access to a skilled programmer.
5. A link to a running marimo instance, e.g. on huggingface (see <https://docs.marimo.io/guides/deploying/#deploy>).
6. A link to a 4-6 minute youtube video in which you (a) demonstrate the functionality of the custom visualisations you created, and (b) how this helps in answering the questions. Paste the full URL in the document instead of a link that automatically opens the web browser (i.e., <https://www.youtube.com/watch?v=DLzxrzFCyOs> instead of [our video](#)). For instructions on how to upload a video, see e.g. <https://www.youtube.com/watch?v=VtF2AgFSLAw>. Important: set visibility to either "public" or "unlisted" (not "private").

Note: It is not forbidden to edit the video. You may include a title screen, but in that case make sure to not lose too much of your limited time doing that, because it is not the primary content.

Intended design (left) vs actual implementation (right). Taken from shirleywu.studio/filmflowers



Max 5 pages.

Part 5. Findings

In this part, describe any interesting/unexpected patterns or signals that you identified using your visualisations. Basically answer the question "what interesting things can you tell about the data"? You can use a story-telling approach: which initial patterns did you find when you started on the project using the question(s) that you defined in Part 2 point 3, how did that lead to focussing on specific parts in the data, etc.

For each finding, please include screenshots/images (if necessary/useful: annotated) that illustrate the patterns/insights that you found.

Max 2 pages.
